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Research Topic: *In silico* identification of repurposed, FDA-approved non-oncology compounds for the Mesenchymal (M) subtype of Triple-Negative Breast Cancer (TNBC) via computational transcriptomic reversal of Wnt and TGF-β signaling

Title: *In silico* identification of repurposed, FDA-approved non-oncology compounds for the Mesenchymal (M) subtype of Triple-Negative Breast Cancer (TNBC) via computational transcriptomic reversal of Wnt and TGF-β signaling

Research Problem

Triple-negative breast cancer (TNBC) is a highly aggressive receptor-null phenotype defined by the absence of estrogen receptor (ER), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2), resulting in a lack of targeted therapeutic options and high patient mortality rates [1]. Among the heterogeneous groups classified by transcriptional profiling, the mesenchymal (M) subtype poses a critical threat due to its intrinsic chemoresistance. Standard neoadjuvant chemotherapy regimens yield a pathologic complete response (pCR) rate of only 31% in this cohort—among the lowest across all TNBC subtypes [2]. Failure to achieve pCR strongly correlates with early tumor recurrence and poor overall survival. This profound treatment failure highlights the need for mesenchymal subtype-specific therapeutic strategies.

Mechanistically, this aggressive subtype is driven by the convergence between TGF-β, a master signaling cytokine that triggers epithelial-mesenchymal transition (EMT), and Wnt/β-catenin signaling, which maintains a robust cancer stem cell (CSC) population and promotes tumor growth [3], [4]. Furthermore, cytotoxic chemotherapy can paradoxically activate the Wnt stemness pathways and drive further resistance to future treatments [1], [4].

Currently, there are no approved therapies specifically engineered to dismantle the unique pathway dependencies of the M subtype. Traditional *de novo* drug discovery takes 10 to 15 years and $1 to $2 billion per approved molecule [5]. To bypass this bottleneck, computationally repurposing existing, safe, FDA-approved non-oncology medications provides an immediate and viable strategy to identify drugs that reverse the subtype's transcriptomic signature and dismantle these dual mechanisms [6].

Existing Literature

Current literature clearly shows that Wnt and TGF-β pathways are the primary drivers for metastasis and chemoresistance in TNBC [3], [4]. To date, current efforts are limited to developing novel inhibitors to block these specific signals. However, the *de novo* compounds frequently fail in human clinical trials, as they are highly toxic to the rest of the body. Furthermore, clinical trials often treat TNBC as a homogenous disease, rather than tailoring the treatment to focus on the M subtype [2].

To overcome these barriers, previous literature has established *in silico* transcriptomic reversal—a method that uses large-scale databases to identify existing drugs capable of reversing gene signatures of a complex disease—as a highly suitable approach for accelerating drug discovery [7]. Complementing this computational tactic, extensive databases such as the Repurposing Drugs in Oncology (ReDO) project have systematically cataloged a wide range of safe, non-cancer medications with documented anti-cancer properties [6]. Combining the computational tools along with repurposing databases creates a highly effective, data-driven strategy to accelerate the discovery of new treatments.

Despite the proven success of computational drug repurposing and a clear understanding of the M subtype and its biological pathways, a research gap remains. No study has applied *in silico* transcriptomic reversal to FDA-approved non-oncology compounds for the simultaneous interception of the Wnt and TGF-β pathways in TNBC. The primary aim of this proposal is to fill this exact gap by applying the computational tools to the M subtype's unique transcriptomic signature, ultimately identifying immediate therapeutic candidates for TNBC.

Research Questions

Primary Research Question

To what extent can safe, FDA-approved non-oncology drugs be computationally repurposed to reverse the transcriptomic signature of mesenchymal triple-negative breast cancer (TNBC) and simultaneously downregulate its driving oncogenic pathways?

Sub-Questions

1. What are the differentially expressed genes (DEGs) that characterize the distinct transcriptomic profile of the mesenchymal TNBC subtype in clinical patient cohorts?
2. Which approved, non-oncology compounds achieve a highly significant transcriptomic reversal score of $\tau \leq -90$, against the mesenchymal signature within the Connectivity Map (CMap) database?
3. Among the top-scoring candidate drugs, which ones successfully target and downregulate the cooperative Wnt/β-catenin and TGF-β signaling axes?

Methodology

Study Design & Variables

This research employs a quantitative, *in silico* retrospective study design.

- **Independent Variable:** Small-molecule drug perturbagen signatures [7]
- **Dependent Variables:** The resulting connectivity score (τ) [8] and the measurable expression changes of key Wnt and TGF-β target genes
- **Controlled Variables:** RNA-seq normalization methods and baseline human cell line lineages

Cohort Selection and Data Acquisition

Clinical genomic data for this study will be acquired from The Cancer Genome Atlas (TCGA) Breast Invasive Carcinoma dataset [9], accessed via the cBioPortal for Cancer Genomics [10]. To isolate the distinct transcriptomic signature of this malignancy, the cohort will be strictly defined by comparing primary tumor samples from patients diagnosed with the M subtype of TNBC [2] against normal solid tissue controls.

Differential Gene Expression Analysis (DGEA)

To isolate the distinct transcriptomic blueprint of Mesenchymal TNBC [2], a differential gene expression analysis (DGEA) will be executed using the DESeq2 software [11] within the R statistical computing environment [12]. To ensure biological relevance and eliminate background noise, strict statistical thresholds will be applied.

- **Upregulated Genes:** $Log_2FC > 1.5$ $(p < 0.05)$
- **Downregulated Genes:** $Log_2FC < -1.5$ $(p < 0.05)$
- **Final Output:** The top 150 upregulated genes and the top 150 downregulated genes of the cancer will be extracted to define the final 300-gene disease signature.

Transcriptomic Reversal Query & Filtering

To identify compounds capable of reversing the pathogenic disease state, the 300-gene M subtype of TNBC signature will be used to computationally screen for therapeutic candidates.

- **Method Used**: Connectivity Map (CMap) [7] L1000 database via the clue.io platform.
- **Efficacy Scoring**: $\tau \leq -90$ [8] is the baseline score for compounds demonstrating highly significant transcriptomic reversal.
- **Safety & Repurposing Filter**: The resulting high-scoring candidates will be cross-referenced against DrugBank 5.0 [13] to ensure only FDA-approved, non-oncology therapeutics are isolated for rapid repurposing [6].

Pathway Enrichment Analysis

To biologically validate the top-scoring drug candidates, it is critical to confirm that their transcriptomic reversal specifically targets the primary drivers of mesenchymal TNBC.

- **Method Used**: Gene set enrichment analysis (GSEA) [14] will be conducted on the gene targets utilizing the clusterProfiler [15] software along with the R programming language [12].
- **Target Validation**: The candidate drugs will be evaluated based on their ability to downregulate the Wnt and TGF-β pathways [3], [4].

Ethical Considerations

This study is entirely *in silico* and does not involve any direct interactions with human subjects or animal models. All clinical data used in this project are sourced from the public domain [9]. All utilized dataset samples are fully de-identified and protected health information (PHI) has been permanently removed prior to public access. Therefore, this secondary data

analysis qualifies for exemption from formal institutional review board (IRB) oversight under standard ethical guidelines.

Potential Limitations

While this *in silico* method enables faster drug discovery, there are also some limitations which must be acknowledged.

- **CMap L1000 Inference Constraints**: The CMap database measures ~1,000 "landmark" genes and infers the remaining genome [7], [8]. Errors could possibly exist in the transcriptomic changes.

- **The Need for Wet-Lab Validation**: A high negative connectivity score (τ) [8] only demonstrates transcriptomic reversal but does not guarantee physical clinical efficacy. The results only act as a filter and the identified drug candidates would ultimately require *in vitro* validation on living TNBC cell lines to prove that the drugs inhibit the Wnt and TGF-β pathways [3], [4].

Expected Outcomes

Upon completion, this *in silico* study is expected to yield:

- A specific 300-gene transcriptomic signature that characterizes the M subtype of TNBC.

- A prioritized list of FDA-approved, non-oncology drugs that demonstrate significant transcriptomic reversal of the M subtype (indicated by a strong negative τ score).

- Confirmation via GSEA that these identified drug candidates successfully target and downregulate the Wnt and TGF-β signaling pathways.

Ultimately, this research will provide a list of low-cost, non-oncology candidates for immediate in vitro testing on TNBC cell lines.

Research Topic

18 TNBC is one of the most clinically lethal breast cancers due to its lack of receptors and high recurrence rates. The M subtype represents an exceptionally urgent therapeutic challenge due to its resistance to chemotherapy and poor overall survival [2]. Despite these clinical needs, there are currently no safe, FDA-approved targeted treatments for the M subtype. This is mainly 24 driven by the crosstalk of the Wnt and TGF-β signaling pathways [3], [4]. Addressing this aggressive cancer requires a shift from the conventional treatments toward an innovative, low-cost alternative by using *in silico* transcriptomic reversal.

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